

( Bonus 2 - řešení )

①  $y$  ... velikost populace

$$\frac{dy}{dt} = k \cdot y - 250 \quad , \quad y(0) = 1800$$

②  $\lambda^2 - 7\lambda + 10 = 0$

$$(\lambda - 2)(\lambda - 5) = 0$$

$$\lambda_1 = 2, \lambda_2 = 5$$

$$y = c_1 \cdot e^{2x} + c_2 \cdot e^{5x}$$

③  $y_p = (ax^2 + bx + c) \cdot e^{5x} \cdot x$

$$y_p = (ax^3 + bx^2 + cx) \cdot e^{5x}$$